
Summary of Part I

This chapter provided a concise introduction to Gaussian probabilistic inference. Here is a brief summary of the key results from this chapter:

- ⊖ Gaussian densities provide the link between probabilistic inference and linear algebra. Though of limited expressiveness, they thus form the basis for computationally efficient inference.
- ⊖ Using linear models, the Gaussian framework can be leveraged for inference on infinite-dimensional objects like functions and curves.
- ⊖ Gaussian processes provide Gaussian inference on infinite-dimensional hypothesis-spaces. Even these nonparametric methods, however, embody non-trivial prior constraints. Like all consistent probabilistic models, have to spread finite probability mass over their hypothesis space.
- ⊖ Gauss–Markov processes, captured by *state-space models*, allow Gaussian inference in linear time, through the algorithms known as filtering and smoothing. They are well-suited for the design of computationally lightweight inference rules, as required in numerical computation. Linear stochastic differential equations capture the continuous-time limit of these stochastic processes.
- ⊖ The parameters of Gaussian models can be inferred using *hierarchical inference*. In most cases this poses a nonlinear (non-Gaussian) optimisation or inference problem. But in the special case of scale and mean of a Gaussian model, *conjugate priors* allow for analytic inference.

The following chapters will employ this toolset in concrete numerical tasks.